EXPERT INSIGHT

In color

Building AI Agents with LLMs, RAG, and Knowledge Graphs

A practical guide to autonomous and modern AI agents

Salvatore Raieli | Gabriele Iuculano

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To Dorotea, Maria, Vincenzo, and Chiara, with love. A small thank you for the immense support.

– Salvatore Raieli

To Marta, for your strength when mine wavered, and for your light in difficult times. Thank you for walking with me through the storms. This book echoes the path we walked together.

– Gabriele Iuculano

The author acknowledges the use of cutting-edge AI, in this case, ChatGPT and Grammarly, with the sole aim of enhancing the language and clarity within the book, thereby ensuring a smooth reading experience for readers. It's important to note that the content itself has been crafted by the author and edited by a professional publishing team.

Contributors

About the authors

Salvatore Raieli is a senior data scientist in a pharmaceutical company with a focus on using AI for drug discovery against cancer. He has led different multidisciplinary projects with LLMs, agents, NLP, and other AI techniques. He has an MSc in AI and a PhD in immunology and has experience in building neural networks to solve complex problems with large datasets. He enjoys building AI applications for concrete challenges that can lead to societal benefits. In his spare time, he writes on his popularization blog on AI (on Medium).

Gabriele Iuculano boasts extensive expertise in embedded systems and AI. Leading a team as the test platform architect, Gabriele has been instrumental in architecting a sophisticated simulation system that underpins a cutting-edge test automation platform.

He is committed to integrating AI-driven solutions, focusing on predictive maintenance systems to anticipate needs and prevent downtimes. He obtained his MSc in AI from the University of Leeds, demonstrating expertise in leveraging AI for system efficiencies. Gabriele aims to revolutionize current business through the power of new disruptive technologies such as AI.

About the reviewers

Malhar Deshpande serves as the director and principal product owner of the AI Center of Excellence at Clean Harbors, where he leads AI initiatives, blending data science, machine learning, and generative AI to transform environmental services. With expertise in technology, innovation, and extensive experience in building AI teams, Malhar Deshpande is recognized for driving innovative solutions. He holds a Bachelor of Engineering, a master’s in information systems, and an MBA from Northeastern University. As a technical reviewer, he is honored to contribute to this book, the AI and technology community, and the future of AI.

I am grateful to my parents, Mohan and Asha Deshpande, for their unwavering support and focus on education. Thanks to my wife, Shruti; my daughter, Tara; and my brother, Dr. Rupak, his wife, Dr. Riteeka, and their daughter, Samaira, for their love and encouragement throughout this journey.

Lalit Chourey is a seasoned software engineer with over a decade of experience in developing scalable backend services and distributed systems, specializing in AI infrastructure for LLM training. Currently a software engineer at Meta Platforms, Lalit leads a team in architecting robust systems for machine learning training. Previously at Microsoft, he led the development of several large-scale cloud services on Azure. Lalit holds a BTech in information technology from the National Institute of Technology, Bhopal, India.

Table of Contents

Introduction to autonomous agents

Toolformer

HuggingGPT

ChemCrow

SwiftDossier

ChemAgent